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Effect of Titrating Positive End-Expiratory Pressure (PEEP) With an Esophageal Pressure–Guided Strategy vs an Empirical High PEEP-FIO_2 Strategy on Death and Days Free From Mechanical Ventilation Among Patients With Acute Respiratory Distress Syndrome

A Randomized Clinical Trial

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IMPORTANCE Adjusting positive end-expiratory pressure (PEEP) to offset pleural pressure might attenuate lung injury and improve patient outcomes in acute respiratory distress syndrome (ARDS).

OBJECTIVE To determine whether PEEP titration guided by esophageal pressure (P_{ES}), an estimate of pleural pressure, was more effective than empirical high PEEP–fraction of inspired oxygen (FIO_2) in moderate to severe ARDS.

DESIGN, SETTING, AND PARTICIPANTS Phase 2 randomized clinical trial conducted at 14 hospitals in North America. Two hundred mechanically ventilated patients aged 16 years and older with moderate to severe ARDS (PaO_2:FIO_2 ≤200 mm Hg) were enrolled between October 31, 2012, and September 14, 2017; long-term follow-up was completed July 30, 2018.

INTERVENTIONS Participants were randomized to P_{ES}-guided PEEP (n = 102) or empirical high PEEP-FIO_2 (n = 98). All participants received low tidal volumes.

MAIN OUTCOMES AND MEASURES The primary outcome was a ranked composite score incorporating death and days free from mechanical ventilation among survivors through day 28. Prespecified secondary outcomes included 28-day mortality, days free from mechanical ventilation among survivors, and need for rescue therapy.

RESULTS Two hundred patients were enrolled (mean [SD] age, 56 [16] years; 46% female) and completed 28-day follow-up. The primary composite end point was not significantly different between treatment groups (probability of more favorable outcome with P_{ES}-guided PEEP: 49.6% [95% CI, 41.7% to 57.5%]; P = .92). At 28 days, 33 of 102 patients (32.4%) assigned to P_{ES}-guided PEEP and 30 of 98 patients (30.6%) assigned to empirical PEEP-FIO_2 died (risk difference, 1.7% [95% CI, −11.1% to 14.6%]; P = .88). Days free from mechanical ventilation among survivors was not significantly different (median [interquartile range]: 22 [15-24] vs 21 [16.5-24] days; median difference, 0 [95% CI, −1 to 2] days; P = .85). Patients assigned to P_{ES}-guided PEEP were significantly less likely to receive rescue therapy (4/102 [3.9%] vs 12/98 [12.2%]; risk difference, −8.3% [95% CI, −15.8% to −0.8%]; P = .04). None of the 7 other prespecified secondary clinical end points were significantly different. Adverse events included gross barotrauma, which occurred in 6 patients with P_{ES}-guided PEEP and 5 patients with empirical PEEP-FIO_2.

CONCLUSIONS AND RELEVANCE Among patients with moderate to severe ARDS, P_{ES}-guided PEEP, compared with empirical high PEEP-FIO_2, resulted in no significant difference in death and days free from mechanical ventilation. These findings do not support P_{ES}-guided PEEP titration in ARDS.

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Limiting mechanical lung injury during invasive ventilation is the most widely accepted approach to improve survival from acute respiratory distress syndrome (ARDS).[1] While lower tidal volumes (V_T) have proven beneficial,[2,3] the clinical benefit, if any, for titrating positive end-expiratory pressure (PEEP) is unclear.

Airway pressure alone may be insufficient to guide lung-protective PEEP titration in ARDS. The chest wall and abdomen contribute unpredictably to pleural pressure and respiratory system mechanics in the critically ill.[4-6] Hence, a given PEEP may facilitate markedly different degrees of lung recruitment and distension in different patients.

Distinguishing lung from chest wall mechanics requires considering transpulmonary pressure (P_L = airway pressure minus pleural pressure), the pressure difference across the lung. Airway pressure is measured readily by modern ventilators, while pleural pressure can be estimated via esophageal manometry using a balloon catheter positioned in the midthoracic esophagus.

In a prior single-center randomized trial (EPVent),[7] esophageal pressure (P_{ES})-guided PEEP titration was compared with the ARDS Network PEEP-fraction of inspired oxygen (FIO_2) table. In that trial, P_{ES}-guided PEEP was associated with higher PaO_2:FIO_2, higher respiratory system compliance, and improved adjusted survival. However, EPVent was conducted at a single center, enrolled patients of any ARDS severity, and used a control strategy with lower PEEP than used in some recent trials[8-12] and suggested by a subsequent meta-analysis for severe ARDS.[13] Therefore, the EPVent-2 trial was conducted to test the hypothesis that adjusting PEEP to achieve a nonnegative P_L is more effective than management using an empirical high PEEP-FIO_2 table.

Key Points

Question What is the clinical benefit of titrating positive end-expiratory pressure (PEEP) according to lung mechanics, delineated using esophageal pressure (P_{ES}), compared with an empirical high PEEP–fraction of inspired oxygen (FIO_2) strategy in the supine patient with moderate to severe acute respiratory distress syndrome (ARDS)?

Finding In this randomized clinical trial that included 200 patients, PEEP titration guided by P_{ES} measurement, compared with empirical high PEEP-FIO_2 titration, resulted in no significant difference in a composite outcome that incorporated death and days free from mechanical ventilation through day 28 (estimated probability of a more favorable outcome with P_{ES}-guided PEEP, 49.6%).

Meaning These findings do not support the use of P_{ES} measurements, instead of an empirical high PEEP-FIO_2 strategy, to guide PEEP titration in patients with moderate to severe ARDS.

Methods

This study was a multicenter randomized trial comparing P_{ES}-guided PEEP titration vs an empirical high PEEP-FIO_2 titration table in patients with moderate to severe ARDS. The trial enrolled participants at 14 hospitals across the United States and Canada. The trial protocol and statistical analysis plan were published previously[14] and are provided in Supplement 1. Each hospital's review board approved the study. Informed consent was provided by study participants or legally authorized surrogates in written form or, at select sites where the local review board permitted, verbally via telephone when written consent was infeasible.

Patients

Patients aged 16 years or older with moderate to severe ARDS (PaO_2:FIO_2 ≤ 200 mm Hg) onset within the last 36 hours were eligible for enrollment. ARDS onset was defined from the moment that all 4 Berlin Definition criteria[15] were met. Patients were excluded for contraindication to esophageal instrumentation or high PEEP, and other reasons enumerated in the eAppendix 1 in Supplement 2 and summarized in **Figure 1**.

Enrolled patients were randomized to either PEEP titration strategy in a 1:1 ratio using a random sorting algorithm with maximum allowable deviation of 6.5% (PASS Sample Size Software, NCSS). Randomization was performed through central web-based software using a balanced randomization scheme by site.

Protocol Elements Common to Both Treatment Groups

After baseline measurements were obtained, a single recruitment maneuver was performed, consisting of a high-pressure breath hold at 35 cm H_2O for 30 seconds. Ventilator settings were then adjusted per protocol (**Table 1**).

P_L was measured during end-inspiratory and end-expiratory breath holds at baseline and at least once daily thereafter on study protocol. Airway and esophageal pressure waveforms were recorded during these measurements and uploaded to a central repository for independent quality control by the core laboratory.

Sedation, neuromuscular blockade, and resuscitation were administered at the discretion of the treating physician. Ventilator weaning was directed by protocol. Patients continued on protocol for 28 days or until breathing unassisted, protocol failure (refractory hypoxemia or acidemia, defined below), withdrawn for safety reasons, rescinded consent, discharged, or death.

P_{ES}-Guided PEEP Group

PEEP was evaluated at least once daily and adjusted as needed to maintain end-expiratory P_L between 0 to 6 cm H_2O, ensuring PEEP was never less than nor substantially more than pleural pressure estimated by P_{ES}. An empirical P_L-FIO_2 table was used (Table 1), targeting the lowest P_L-FIO_2 combination that maintained oxygenation goals.

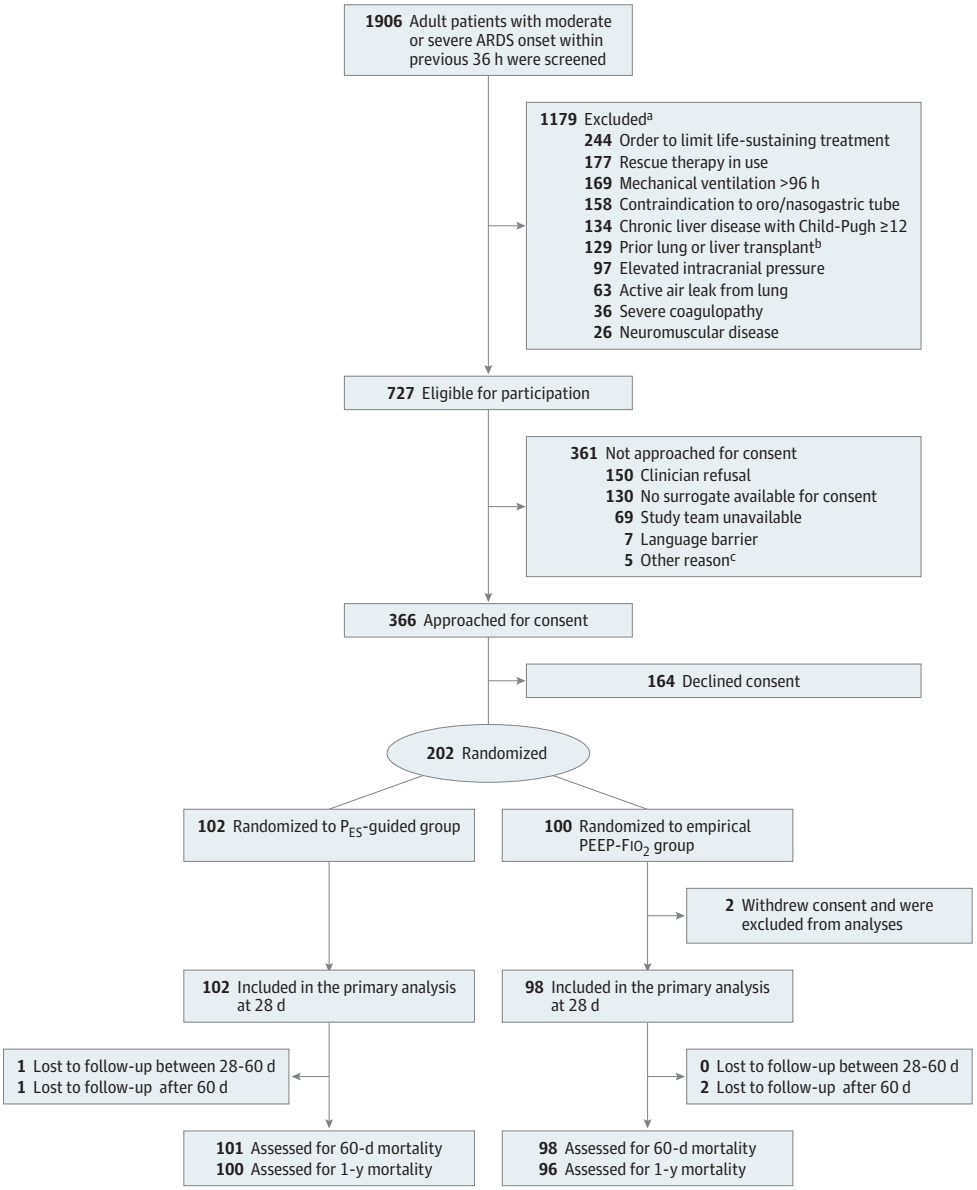
To lessen tidal overdistension, if end-inspiratory P_L exceeded 20 cm H_2O, V_T was decreased to as low as 4 mL/kg predicted body weight (PBW). For severe dyspnea or acidemia, V_T could be increased to as high as 8 mL/kg PBW provided end-inspiratory P_L remained 20 cm H_2O or less.

Once end-expiratory P_L of 0 with FIO_2 of 0.5 or less was tolerated for at least 24 hours, the patient was transitioned to a

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Figure 1. Screening, Enrollment, Randomization, and Follow-up of Study Participants



ARDS indicates acute respiratory distress syndrome.

[a] Detailed eligibility criteria are provided in the eAppendix in Supplement 2. The sum of reasons excluded exceeds the total number of patients excluded because some patients met multiple exclusion criteria.

[b] The initial protocol excluded liver transplant recipients. This exclusion was removed in a protocol amendment during the trial.

[c] Other reasons eligible patients were not approached for consent include death prior to approach (n = 2), extubation prior to approach (n = 1), local site exclusion for pregnancy (n = 1), and reason not reported (n = 1).

weaning protocol and PEEP and FIO_2 were incrementally decreased further without regard for P_L (Supplement 1).

Empirical PEEP-FIO_2 Group

PEEP was adjusted to maintain the lowest PEEP-FIO_2 combination possible on the empirical table while maintaining oxygenation goals (Table 1). The empirical table was adopted from the control group of the recent OSCILLATE trial.[8] Crossover to the P_{ES}-guided strategy was prohibited. P_L measurements were not disclosed to the treating team nor considered in selecting PEEP. To lessen tidal overdistension, V_T could be decreased to as low as 4 mL/kg PBW if airway plateau pressure exceeded 35 cm H_2O, a threshold within the range used in recent ARDS trials.[3,8,10] For severe dyspnea or acidemia, V_T could be increased to as high as 8 mL/kg PBW provided airway plateau pressure remained at 35 cm H_2O or less.

Primary End Point

The prespecified primary end point was a ranked composite score that incorporated death and days free from mechanical

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Table 1. Summary of Ventilator Protocols

Protocol Variable	P_{ES}-Guided PEEP		Empirical PEEP-FIO_2	
Ventilator mode	Volume or pressure assist control		Volume or pressure assist control	
Tidal volume, mL/kg PBW	6 (range, 4-8)		6 (range, 4-8)	
End-inspiratory pressure limit, cm H_2O	$P_L \leq 20$		$P_{PLAT} \leq 35$	
Respiratory rate set to attain target pH 7.30-7.45, breaths/min	6-35		6-35	
Inspiratory to expiratory time ratio	1.1-1.3		1.1-1.3	
Goal oxygenation	PaO_2: 55-80 mm Hg or SpO_2: 88%-93%		PaO_2: 55-80 mm Hg or SpO_2: 88%-93%	
Allowable combinations of FIO_2 and either end-expiratory P_L or PEEP to attain goal oxygenation[a]	FIO_2	P_L, cm H_2O	FIO_2	PEEP, cm H_2O
	0.3	0	0.3	5
	0.4	0	0.3	8
	0.5	0	0.3	10
	0.5	2	0.4	10
	0.6	2	0.4	12
	0.6	3	0.4	14
	0.7	3	0.4	16
	0.7	4	0.4	18
	0.8	4	0.5	18
	0.8	5	0.5	20
	0.9	5	0.6	20
	0.9	6	0.7	20
	1.0	6	0.8	20
			0.8	22
			0.9	22
			1.0	22
			1.0	24
Single recruitment maneuver on enrollment to standardize lung volume history	Yes		Yes	
Subsequent recruitment maneuvers	Only if required to reverse hypoxemia after suction or circuit disconnect		Only if required to reverse hypoxemia after suction or circuit disconnect	
Criteria to initiate ventilator weaning on pressure support mode	$FIO_2 \leq 0.4$, PEEP ≤ 10 cm H_2O, patient able to trigger ventilator, and no hemodynamic instability		$FIO_2 \leq 0.4$, PEEP ≤ 10 cm H_2O, patient able to trigger ventilator, and no hemodynamic instability	

Abbreviations: FIO_2, fraction of inspired oxygen; PaO_2, partial pressure of arterial oxygen; PBW, predicted body weight; PEEP, positive end-expiratory pressure; P_{ES}, esophageal pressure; P_L, transpulmonary pressure (P_L = airway pressure minus pleural pressure); P_{PLAT}, plateau airway pressure; SpO_2, oxygen saturation as measured by pulse oximetry.

[a] In the P_{ES}-guided group, once patients reached P_L of 0 cm H_2O, they were transitioned to empirical PEEP-FIO_2 table at no more than PEEP of 14 cm H_2O and FIO_2 of 0.4. Further PEEP weaning occurred per the empirical table. If hypoxemia occurred, necessitating higher PEEP-FIO_2 combination, patients were returned to P_{ES}-guided PEEP table.

ventilation through day 28, calculated in such a manner that death constitutes a worse outcome than fewer days off the ventilator.[16] Time free from mechanical ventilation was calculated as the number of days between successful liberation from the ventilator and study day 28. Each patient was compared with every other patient in the study and assigned a score (tie: 0, win: +1, loss: −1) for each pairwise comparison based on whom fared better. If one patient survived and the other did not, scores of +1 and −1 were assigned, respectively, for that pairwise comparison. If both patients in the pairwise comparison survived, the assigned score depended on which patient had more days free from mechanical ventilation: the patient with more days off the ventilator received a score of +1, while the patient with fewer days received a score of −1. If both patients survived and had the same number of days off the ventilator, or if both patients died, they both were assigned a score of 0 for that pairwise comparison. For each patient, scores for all pairwise comparisons were summed, resulting in a cumulative score for each patient. These cumulative scores were ranked and compared between treatment groups via the Mann-Whitney technique.

Effect size is reported as the probability of more favorable outcome, also known as the probabilistic index, which describes the estimated probability that an individual randomly selected from one treatment group will have a higher score (more favorable outcome) than an individual randomly selected from the other group.[17,18] The probability of more favorable outcome is mathematically equivalent to the area under the receiver operating characteristic curve for the nonparametric Mann-Whitney U test.[17]

Secondary End Points

Prespecified secondary clinical end points included all-cause mortality at 28 days, 60 days, and 1 year; ventilator-free days through day 28; intensive care unit (ICU) and hospital lengths of stay through day 28 and day 60; protocol failure requiring rescue therapy; and functional status at 1 year. Ventilator-free days were calculated as previously described,[19] assigning

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a value of 0 failure-free days for patients who died before day 28. Protocol failure was defined as refractory hypoxemia (PaO_2 < 55 mm Hg or oxygen saturation as measured by pulse oximetry [SpO_2] < 88%) despite maximum PEEP and FIO_2 settings or refractory acidemia (pH < 7.15) despite maximum V_T and respiratory rate settings permitted per the protocol. Functional outcomes were evaluated at 1 year among survivors using surveys assessing independence with activities of daily living, physical and mental health, and frailty (eAppendix 1 in Supplement 2). Insufficient data were collected to determine ICU length of stay at 60 days. Plasma biomarkers of lung injury, a prespecified secondary end point, are not reported here because results are not yet available. Differences in several respiratory physiological measures were prespecified as secondary mechanistic end points (eAppendix 1 in Supplement 2).

Safety End Points

Prespecified safety end points included shock-free days (vasopressor requirement) through day 28, pneumothorax, bronchopleural fistula, any barotrauma, acute kidney injury requiring renal replacement therapy in the first 28 days, and any other serious adverse event. Shock was defined as receiving any vasopressor or inotropic infusion, and patients transferred out of the ICU were assumed to be shock free; shock-free days otherwise was calculated in a manner similar to ventilator-free days as described above. Barotrauma was defined by the presence of pneumothorax, pneumomediastinum, subcutaneous emphysema, or bronchopleural fistula. Patients receiving long-term dialysis at baseline were excluded from determination of acute kidney injury. Although listed as a safety end point in the original protocol, data on fluid balance were not collected.

Statistical Analyses

Using estimates derived from the prior EPVent trial,[7] a sample of 200 patients was estimated to provide 85% power to detect a significant difference in the primary ranked composite outcome with 2-sided alpha of .05. Sample size calculations assumed 28-day mortality of 30% with empirical PEEP-FIO_2 and 20% with P_{ES}-guided PEEP. The distribution of days free from mechanical ventilation was assigned in 2 stages. First, the proportion of patients with 0 days free from mechanical ventilation was assumed to be 10% with empirical PEEP-FIO_2 and 15% with P_{ES}-guided PEEP, reflecting a larger proportion of patients alive but still receiving mechanical ventilation at day 28 in the P_{ES}-guided PEEP group. Second, remaining nonzero values were assumed to be normally distributed with a mean (SD) of 13 (6.5) and 15 (6.5) days free from mechanical ventilation with empirical PEEP-FIO_2 and P_{ES}-guided PEEP, respectively.

Primary analyses were performed according to randomization group. Patients for whom consent was withdrawn were excluded from all analyses. Descriptive statistics are presented as number (percentage), mean (SD), or median (interquartile range [IQR]). Differences in respiratory physiological measures from days 1 through 7 on study protocol were analyzed using linear mixed models, entering treatment group, time, and group-time interaction as fixed effects with random intercept and unstructured covariance matrix. The main interest in these models was whether the physiological measures differed on average across the first 7 days on protocol, assessed by considering the fixed effect for treatment group. Whether change over time differed by treatment group, assessed by group-time interaction, was also evaluated. A post hoc analysis was performed to evaluate how often initial on-protocol PEEP might have differed by more than 2 cm H_2O, given a patient's observed FIO_2, had that patient been assigned to the other treatment group.

The rank composite score primary end point was compared using the Mann-Whitney U test and effect size described with the probability of more favorable outcome (probabilistic index) as described above; 95% CIs were calculated per the Newcombe method.[20] Survival curves were generated via the Kaplan-Meier method and compared with a log-rank test. The proportional hazards assumption for the log-rank test was assessed by the method of Lin et al[21] and confirmed to be valid. Mortality at 28 days, 60 days, and 1 year; acute kidney injury; pneumothorax; barotrauma; and use of rescue therapy were compared using Fisher exact test; and effect estimates were reported as the absolute risk difference with asymptotic 95% CIs. Days free from mechanical ventilation among survivors, ventilator- and shock-free days, and ICU and hospital lengths of stay were non-normally distributed and therefore compared using the Mann-Whitney U test; effect estimates were reported as the median difference calculated using the unbiased Hodges-Lehmann estimator[22] with corresponding 95% CIs. Analyses of clinical end points were reported using available data with no imputation for missing data given the low observed rate of missingness.

For all analyses, a 2-sided alpha threshold of .05 was considered significant. Because of the potential for type 1 error due to multiple comparisons without adjustment, findings for analyses of secondary end points should be interpreted as exploratory. Analyses were conducted using SAS version 9.3 (SAS Institute Inc).

Results

Study Population

A total of 727 screened patients were eligible for participation, of whom 102 were randomized to the P_{ES}-guided strategy and 100 to the empirical PEEP-FIO_2 strategy between October 31, 2012, and September 14, 2017. Median enrollment across the 14 participating sites was 9 patients (IQR, 2.25-15.5). Consent was withdrawn for 2 patients assigned to the PEEP-FIO_2 strategy prior to protocol initiation, leaving 102 and 98 patients in the P_{ES}-guided and PEEP-FIO_2 groups, respectively, for inclusion in the primary analysis (Figure 1). An esophageal balloon catheter could not be inserted successfully in 1 patient randomized to P_{ES}-guided PEEP; that patient did not undergo subsequent study procedures but regardless did complete 28-day follow-up and was included in analyses of all clinical end points. Baseline characteristics are presented in **Table 2**.

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Table 2. Baseline Characteristics of Study Participants

Variable	Median (IQR)	
	P_{ES}-Guided PEEP (n = 102)	Empirical PEEP-FIO_2 (n = 98)
Age, y	58 (47 to 66)	57.5 (43 to 69)
Sex, No. (%)		
Female	38 (37.3)	54 (55.1)
Male	64 (62.7)	44 (44.9)
PBW, mean (SD), kg	63.6 (11.8)	60.4 (12.2)
Actual body weight, kg	84.0 (72.0 to 105.0)	79.5 (68.2 to 97.9)
APACHE-II score, mean (SD)[a]	27 (8)	28 (7)
SOFA score, mean (SD)[b]	11 (4)	11 (4)
Time intubated prior to enrollment, h	22 (13 to 31)	21 (15 to 29)
ARDS risk factors, No. (%)		
Sepsis	86 (84.3)	85 (86.7)
Pneumonia	71 (69.6)	78 (79.6)
Aspiration	20 (19.6)	22 (22.4)
Prolonged shock	9 (8.8)	17 (17.3)
Multiple transfusions	10 (9.8)	9 (9.2)
Acute pancreatitis	3 (2.9)	4 (4.1)
Trauma	3 (2.9)	3 (3.1)
Any pulmonary risk factor	82 (80.4)	88 (89.8)
Respiratory characteristics		
pH	7.33 (7.27 to 7.37)	7.33 (7.25 to 7.40)
$PaCO_2$, mm Hg	44 (38 to 52)	43 (37 to 51)
PaO_2, mm Hg	71 (61 to 86)	69 (61 to 84)
PaO_2:FIO_2, mm Hg	95 (73 to 129)	90 (69 to 123)
Tidal volume, mL/kg PBW	6.2 (5.9 to 6.7)	6.2 (5.8 to 7.1)
Airway pressure, cm H_2O		
Plateau	28 (24 to 32)	27 (25 to 30)
Mean	20 (16 to 24)	19 (16 to 22)
Set PEEP, cm H_2O	14 (10 to 18)	12.5 (10 to 16)
FIO_2	0.60 (0.50 to 0.80)	0.60 (0.50 to 0.70)
Respiratory rate, breaths/min	26 (22 to 30)	26 (22 to 30)
Minute ventilation, L/min	10.3 (8.5 to 12.1)	9.4 (8.4 to 11.3)
P_{ES}, cm H_2O		
At end-inspiration	19 (16 to 21)	18 (16 to 21)
At end-expiration	16 (13 to 19)	15 (13 to 18)
P_L, cm H_2O		
At end-inspiration	8 (6 to 11)	9 (5 to 12)
At end-expiration	0 (−3 to 1)	−1 (−3 to 2)
Airway driving pressure, cm H_2O	13 (10 to 15)	13 (11 to 15)
Transpulmonary driving pressure, cm H_2O	8 (7 to 11)	9 (7 to 11)
Lung compliance, mL/cm H_2O	48 (34 to 63)	42 (33 to 55)
Respiratory system compliance, mL/cm H_2O	32 (25 to 41)	30 (23 to 37)
Cointerventions at baseline, No. (%)		
Neuromuscular blockade	32 (31.4)	34 (34.7)
Vasopressors or inotropes	59 (57.8)	55 (56.1)
Systemic corticosteroids	33 (32.4)	36 (36.7)

Abbreviations: ARDS, acute respiratory distress syndrome; FIO_2, fraction of inspired oxygen; IQR, interquartile range; $PaCO_2$, partial pressure of arterial carbon dioxide; PaO_2, partial pressure of arterial oxygen; PBW, predicted body weight; PEEP, positive end-expiratory pressure; P_{ES}, esophageal pressure; P_L, transpulmonary pressure; SOFA, Sequential Organ Failure Assessment.

[a] APACHE-II ranges from 0 to 71; higher scores indicate greater illness severity. A score of 25 to 29 predicts a mortality risk of approximately 35% to 50%.

[b] SOFA score ranges from 0 to 24; higher scores indicate greater illness severity. A score of 10 to 12 predicts a mortality risk of approximately 40% to 50%.

Ventilator Management and Respiratory Physiology

Protocol adherence was high: 93.3% of observed PEEP values in the P_{ES}-guided PEEP group and 94.6% of values in the empirical PEEP-FIO_2 group were within ±2 cm H_2O of the protocol-specified value for the observed FIO_2 (eTable 1 in Supplement 2). Agreement between site and physiology core laboratory readings of P_{ES} and P_L also was high (eAppendix 2 in Supplement 2).

On initiating protocol ventilator settings, PEEP changed by a mean (SD) of +3 (6) cm H_2O in the P_{ES}-guided group and +3 (4) cm H_2O in the empirical PEEP-FIO_2 group (P = .89). The proportion of patients in whom PEEP changed (increase

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or decrease) by at least 5 cm H_2O was not significantly different between treatment groups (37.6% for P_{ES}-guided PEEP and 35.7% for empirical PEEP-FIO_2; P = .88).

In the P_{ES}-guided group, PEEP was increased by as much as 20 cm H_2O and decreased by as much as 12 cm H_2O from baseline to first values on protocol. With initiation of the protocol, PEEP was increased in 64.3%, decreased in 16.8%, and unchanged in 18.8% of patients assigned to P_{ES}-guided PEEP.

By comparison, in the empirical PEEP-FIO_2 group, PEEP was increased by as much as 13 cm H_2O and decreased by as much as 5 cm H_2O from baseline to first values on protocol. With initiation of the protocol, PEEP was increased in 57.1%, decreased in 15.3%, and unchanged in 27.6% of patients assigned to empirical PEEP-FIO_2.

In the P_{ES}-guided group, the highest PEEP on protocol was 36 cm H_2O; 12 patients had at least 1 day where PEEP exceeded 24 cm H_2O. The empirical PEEP-FIO_2 group limited maximum PEEP to 24 cm H_2O. The mean (SD) PEEP and FIO_2 the first day on protocol were 17 (6) cm H_2O and 0.56 (0.15), respectively, in the P_{ES}-guided group vs 16 (4) cm H_2O and 0.51 (0.17) in the empirical PEEP-FiO_2 group (P = .28 for PEEP; P = .048 for FIO_2).

There were no significant differences in end-expiratory P_L on average over the first 7 days, although the rate of decline in end-expiratory P_L from day 1 to day 7 was significantly slower in patients assigned to P_{ES}-guided PEEP (**Figure 2**). Values for end-expiratory P_{ES}, PEEP, end-inspiratory P_L, plateau pressure, transpulmonary driving pressure, and PaO_2:FIO_2 were not significantly different between treatment groups over the first 7 days (eTable 2 in Supplement 2).

Clinical End Points

Complete data for clinical end points were available for all study participants except for 60-day mortality (1 patient assigned to P_{ES}-guided PEEP was lost to follow-up), 1-year mortality (2 patients from each group were lost to follow-up), and 1-year survey-derived functional outcomes among survivors (eAppendix 2 in Supplement 2).

Primary End Point

The primary ranked composite end point, incorporating death and days free from mechanical ventilation through day 28, was not significantly different between treatment groups (**Table 3**). The probability of more favorable outcome with P_{ES}-guided PEEP, compared with empirical PEEP-FIO_2, was 49.6% (95% CI, 41.7% to 57.5%; P = .92).

Secondary End Points

Thirty-three patients (32.4%) assigned to P_{ES}-guided PEEP and 30 patients (30.6%) assigned to empirical PEEP-FIO_2 died by study day 28 (risk difference, 1.7% [95% CI, −11.1% to 14.6%]; P = .88). Neither 60-day mortality (37.6% vs 37.8%; risk difference, −0.1% [95% CI, −13.6% to 13.3%]; P > .99) nor 1-year mortality (44.0% vs 45.8%; risk difference, −1.8% [95% CI, −15.8% to 12.1%]; P = .89) was significantly different between groups (Table 3; **Figure 3**).

Ventilator-free days was not significantly different between treatment groups (P_{ES}-guided PEEP: median, 15.5 [IQR, 0-23] days, and empirical PEEP-FIO_2: median 17.5 [IQR, 0-23] days; median difference, 0 [95% CI, 0 to 0] days; P = .93).

Sixteen patients received protocol-defined rescue therapy during the trial, initiated a median of 4 (IQR, 2-6) days after enrollment. Rescue was initiated for refractory hypoxemia in 10 cases (62.5%), refractory acidemia in 2 cases (12.5%), and reasons not reported in 4 cases (25.0%). Patients assigned to P_{ES}-guided PEEP were significantly less likely to receive any rescue therapy than patients assigned to empirical PEEP-FIO_2 (3.9% vs 12.2%; risk difference, −8.3% [95% CI, −15.8% to −0.8%]; P = .04). Significantly fewer patients assigned to P_{ES}-guided PEEP received inhaled pulmonary vasodilators (2.9% vs 10.2%; risk difference, −7.3% [95% CI, −14.1% to −0.4%]; P = .046). Differences in use of rescue prone positioning (1.0% vs 3.1%; risk difference, −2.1% [95% CI, −6.0% to 1.8%]; P = .36) and extracorporeal membrane oxygenation (1.0% vs 3.1%; risk difference, −2.1% [95% CI, −6.0% to 1.8%]; P = .36) did not achieve statistical significance. Additional characteristics of patients receiving rescue therapy are presented in eTables 3-5 in Supplement 2.

Other secondary end points are presented in Table 3 and eTable 6 in Supplement 2.

Safety End Points

Shock-free days through day 28 did not significantly differ between patients assigned to P_{ES}-guided PEEP vs empirical PEEP-FIO_2 (median, 14 [IQR, 0-21] vs 17 [IQR, 0-21] days; median difference, 0 [95% CI, −2 to 0] days; P = .47; Table 3). Hemodynamic instability was reported as a serious adverse event for 1 patient in the P_{ES}-guided PEEP group on protocol initiation. There were no study-related cardiac arrests in either group nor other reported study-related serious adverse events. No complications related to esophageal balloon catheter insertion were reported.

Incidence of pneumothorax (2.9% vs 2.0%; risk difference, 0.9% [95% CI, −3.4% to 5.2%]; P > .99) and any barotrauma (5.9% vs 5.1%; risk difference, 0.8% [95% CI, −5.5% to 7.1%]; P > .99) also were not significantly different between treatment groups. No patients in either treatment group developed a bronchopleural fistula.

Acute kidney injury requiring initiation of renal replacement therapy in the first 28 days occurred in 21.0% of patients assigned to P_{ES}-guided PEEP vs 33.3% assigned to empirical PEEP-FIO_2 (risk difference, −12.3% [95% CI, −24.7% to 0.0%]; P = .056).

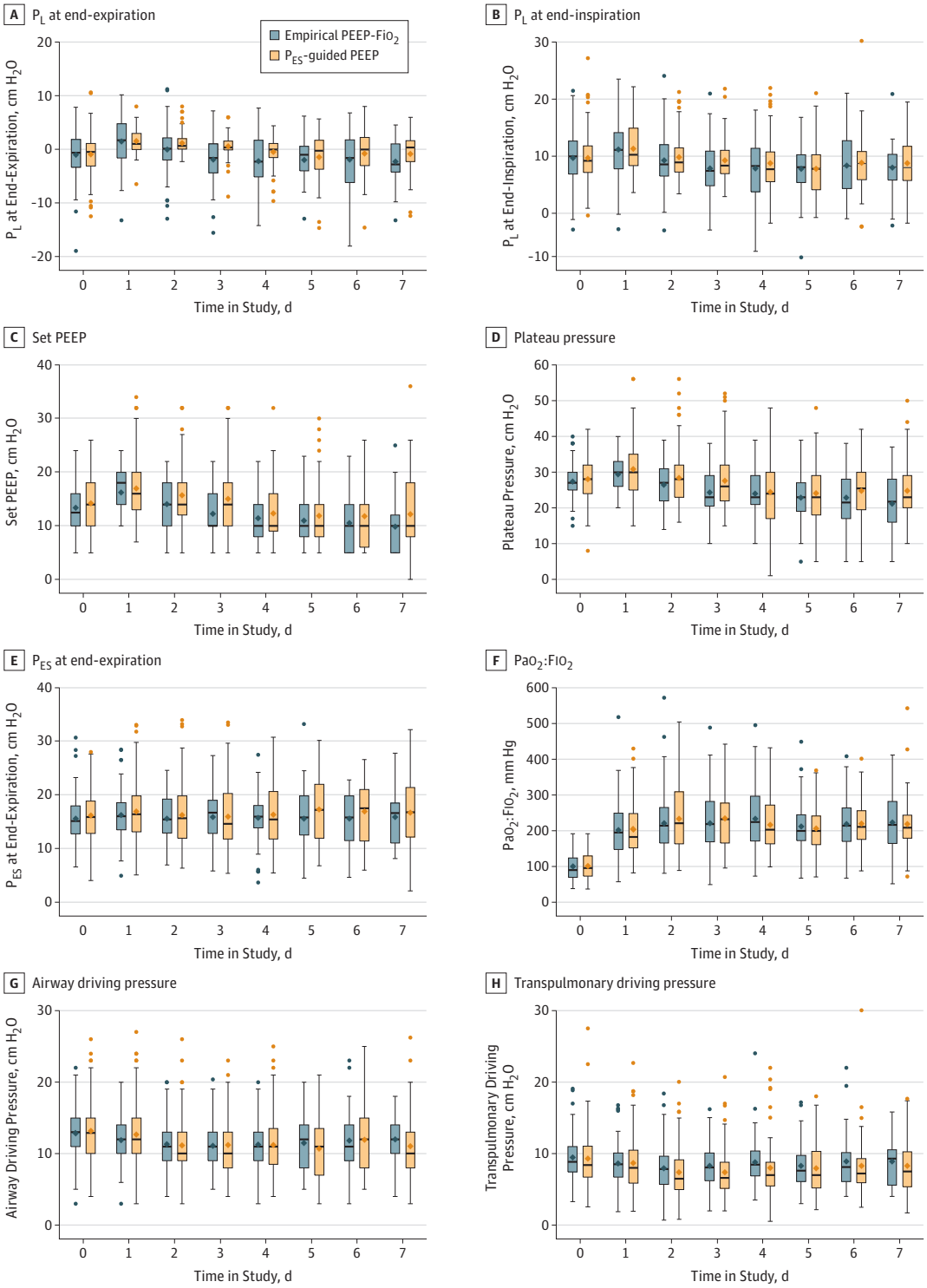
Post Hoc Analysis

A post hoc analysis evaluated how frequently initial on-protocol PEEP values might have differed, given the FIO_2 requirement, had patients been assigned to the other treatment group. Of 192 patients with requisite data for this determination, initial on-protocol PEEP would have differed by more than 2 cm H_2O in 103 cases (53.6%) if assigned to the other treatment group.

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Figure 2. Respiratory Physiological Measures From Baseline Through Day 7



Boxes represent median and interquartile range. Whiskers extend 1.5 times the interquartile range beyond the first and third quartiles per the conventional Tukey method. Circles beyond the whiskers represent outliers. Diamonds represent mean values. Day 0 denotes baseline preintervention values. The number of patients with available respiratory physiological data decreases over successive study days due to deaths and discontinuation of invasive mechanical ventilation. Transpulmonary pressure (P_L) equals airway pressure minus pleural pressure. Airway driving pressure equals plateau pressure minus positive end-expiratory pressure (PEEP). Transpulmonary driving pressure equals end-inspiratory P_L minus end-expiratory P_L. FIO_2 indicates fraction of inspired oxygen; PaO_2, partial pressure of arterial oxygen; and P_{ES}, esophageal pressure.

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Table 3. Patient Outcomes[a]

Variable	P_{ES}-Guided PEEP (n = 102)	Empirical PEEP-FIO_2 (n = 98)	Absolute Difference, % (95% CI)[b]	*P* Value[c]
Primary End Point				
Probability of more favorable outcome, a ranked composite incorporating death and days free from mechanical ventilation among survivors, % (95% CI)[d]	49.6 (41.7 to 57.5)	50.4 (42.5 to 58.3)	NR[e]	.92
Secondary Clinical End Points				
Mortality through day 28, No. (%)	33 (32.4)	30 (30.6)	1.7 (−11.1 to 14.6)	.88
Days free from mechanical ventilation among survivors through day 28, median (IQR)	22 (15 to 24)	21 (16.5 to 24)	0 (−1 to 2)	.85
Mortality through day 60, No./total No. (%)	38/101 (37.6)	37/98 (37.8)	−0.1 (−13.6 to 13.3)	>.99
Mortality through 1 y, No./total No. (%)	44/100 (44.0)	44/96 (45.8)	−1.8 (−15.8 to 12.1)	.89
Ventilator-free days through day 28, median (IQR)[f]	15.5 (0 to 23)	17.5 (0 to 23)	0 (0 to 0)	.93
ICU length of stay through day 28, median (IQR), d	10 (6 to 17)	9.5 (5 to 14)	1 (−1 to 3)	.24
Hospital length of stay through day 28, median (IQR), d	16 (9 to 26)	15 (8 to 24)	0 (−1 to 3)	.58
Hospital length of stay through day 60, median (IQR), d	16 (9 to 26)	15 (8 to 24)	1 (−2 to 4)	.47
Rescue therapy administered, No. (%)[g]	4 (3.9)	12 (12.2)	−8.3 (−15.8 to -0.8)	.04
Prone positioning, No. (%)	1 (1.0)	3 (3.1)	−2.1 (−6.0 to 1.8)	.36
Inhaled pulmonary vasodilator, No. (%)	3 (2.9)	10 (10.2)	−7.3 (−14.1 to −0.4)	.046
Extracorporeal membrane oxygenation, No. (%)	1 (1.0)	3 (3.1)	−2.1 (−6.0 to 1.8)	.36
Recruitment maneuvers, No. (%)	1 (1.0)	1 (1.0)	0.0 (−2.8 to 2.7)	>.99
Safety End Points				
Shock-free days, median (IQR)[f]	14 (0 to 21)	17 (0 to 21)	0 (−2 to 0)	.47
Acute kidney injury requiring renal replacement therapy in the first 28 d, No./total, No. (%)[h]	21/100 (21.0)	32/96 (33.3)	−12.3 (−24.7 to 0.0)	.056
Pneumothorax, No. (%)	3 (2.9)	2 (2.0)	0.9 (−3.4 to 5.2)	>.99
Bronchopleural fistula, No.	0	0	0	
Barotrauma, No. (%)[i]	6 (5.9)	5 (5.1)	0.8 (−5.5 to 7.1)	>.99

Abbreviations: ICU, intensive care unit; IQR, interquartile range; NR, not reported; PEEP, positive end-expiratory pressure; P_{ES}, esophageal pressure.

[a] Complete outcome data were available for all study participants except for 60-day mortality, for which 1 patient assigned to P_{ES}-guided PEEP was lost to follow-up, and 1-year mortality, for which 2 patients from each group were lost to follow-up.

[b] For variables reported as median (IQR), Hodges-Lehmann estimation was used to determine unbiased median differences and their 95% CIs. The result represents the median difference among all possible pairs formed by 1 patient from each group and thus does not necessarily equal the crude difference between group medians.

[c] Multiple testing was performed without adjustment, such that analyses of secondary end points should be considered exploratory.

[d] Probability of more favorable outcome, also known as the probabilistic index, is the estimated probability that an individual randomly selected from one treatment group will have a higher score (more favorable outcome) than an individual randomly selected from the other group. It is mathematically equivalent to the area under the receiver operating characteristic curve for the Mann-Whitney U test. A value of 50% indicates no difference between groups.

[e] No absolute difference is reported for the primary end point effect estimate, probability of more favorable outcome, because the probability for either group is itself the comparator effect estimate; their combined probability equals 100%.

[f] Ventilator-free days and shock-free days through day 28 are composite end points that assign a value of 0 free days for patients who died before day 28.

[g] Protocol-defined rescue therapies included prone positioning, inhaled pulmonary vasodilators (eg, nitric oxide, epoprostenol), high-frequency oscillation, airway pressure release ventilation, and extracorporeal membrane oxygenation. No patients received rescue high-frequency oscillation or airway pressure release ventilation.

[h] Two patients in each group were on long-term dialysis at baseline and excluded from calculation of acute kidney injury.

[i] Barotrauma includes pneumothorax, pneumomediastinum, or subcutaneous emphysema.

Discussion

In this trial comparing P_{ES}-guided PEEP with empirical high PEEP-FIO_2 in patients with moderate to severe ARDS, there was no significant difference in the primary composite end point incorporating death and days free from mechanical ventilation through day 28.

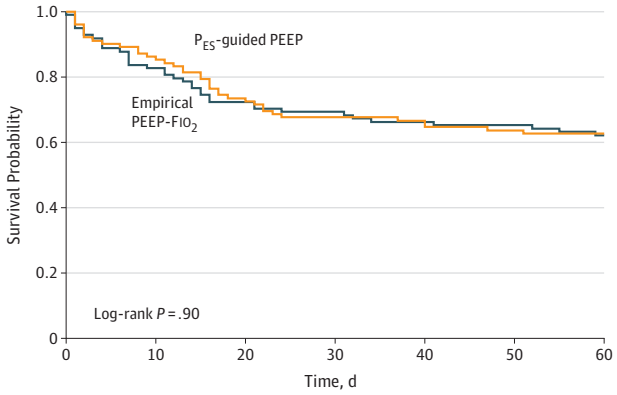
Repetitive opening and closing of collapsed but recruitable lung units during tidal ventilation generates high local shear forces that cause alveolar injury, a phenomenon termed *atelectrauma*.[1,23,24] Thus, in theory, PEEP should be set high enough to keep open most lung units at risk of atelectrauma and decrease regional mechanical heterogeneity while also sufficiently low to minimize overdistension of more compliant lung regions. The P_{ES}-guided PEEP strategy in this trial aimed to implement this theory by titrating PEEP to achieve P_L near zero at end-expiration. Yet, the empirical PEEP-FIO_2 protocol unexpectedly resulted in P_L values that on average were not significantly different from the P_{ES}-guided PEEP strategy (Figure 2). Still, knowing a patient's P_{ES} and P_L values may increase clinician comfort with uptitrating PEEP.

Any benefit of preventing atelectrauma with higher PEEP or end-expiratory P_L could be offset by adverse effects from increasing lung volume at end-inspiration, exacerbating tidal overdistension.[25-27] In the recent ART trial,[28] increased barotrauma, hemodynamic instability, and mortality were observed with an open lung strategy (aggressive recruitment

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Figure 3. Kaplan-Meier Survival Analysis Through Day 60



No. at risk

P_{ES}-guided PEEP	102	88	75	68	67	64	63
Empirical PEEP-FIO_2	98	81	71	68	65	64	61

FIO_2 indicates fraction of inspired oxygen; PEEP, positive end-expiratory pressure; and P_{ES}, esophageal pressure.

maneuver plus comparatively high PEEP), compared with a low PEEP strategy, in patients with moderate to severe ARDS. An ideal lung-protective protocol might require lowering V_T further as PEEP or end-expiratory P_L is increased, an untested strategy that warrants consideration. Time to PEEP titration also is an underinvestigated factor. Earlier application of adequate PEEP, within the first few hours after intubation, may be necessary to abate alveolar instability and injury.[29,30]

Several important differences exist between this study and the prior EPVent trial,[7] which observed an adjusted survival benefit with P_{ES}-guided PEEP. First, the comparator group in the prior trial was a less aggressive empirical PEEP-FIO_2 strategy, resulting in significantly different end-expiratory P_L on protocol between groups, unlike the present study where early end-expiratory P_L was not significantly different between treatment groups. Post hoc analysis suggested that the 2 PEEP protocols in this trial would have issued different initial titration instructions in approximately half of patients. Second, the empirical PEEP-FIO_2 comparator strategy in the prior trial resulted in a mean end-expiratory P_L less than 0 cm H_2O at all measured times, likely predisposing to atelectrauma, whereas the empirical PEEP-FIO_2 comparator strategy in the present trial maintained a mean end-expiratory P_L of 0 cm H_2O or greater until day 3. Third, in the prior trial, driving pressure decreased once patients assigned to P_{ES}-guided PEEP were placed on protocol, but no change was observed in the comparator group, suggesting PEEP-associated lung recruitment only in the P_{ES}-guided group.[31] In the present trial, there was no separation between treatment groups in airway or transpulmonary driving pressure, and no meaningful difference from baseline to first values on protocol, suggesting minimal lung recruitment with protocol-driven PEEP. Fourth, most patients in the prior trial had extrapulmonary risk factors for ARDS, including 40% with an identified intra-abdominal risk factor (eg, pancreatitis, cholangitis, bowel obstruction, or perforation), unlike the present study where pulmonary risk factors predominated. ARDS due to intra-abdominal pathology in particular might exhibit greater lung recruitability owing to the contribution of higher intra-abdominal pressures to pleural pressure and associated atelectasis.[32]

Limitations

This study has several limitations. First, it was underpowered for smaller—but still clinically meaningful—differences in survival and other end points, as evident from the wide 95% CIs for several outcomes in Table 3. Second, methodological issues with P_{ES} and P_L measurements could have impeded implementation of the P_{ES}-guided PEEP strategy, although agreement on P_{ES} and P_L readings was high between site investigators and the core laboratory. Third, while balloon position in the retrocardiac midthoracic esophagus affords a reasonable estimate of average pleural pressure in the chest,[5,33-35] spatial pleural pressure gradients may contribute to regional overdistension and cyclic atelectasis.[36]

Fourth, whether either PEEP strategy in this trial is superior (or inferior) to usual care or a less aggressive PEEP-FIO_2 strategy in patients with moderate to severe ARDS remains uncertain. Fifth, this trial protocol prohibited prone positioning except as rescue therapy. A multicenter trial of prone positioning, published while this trial was ongoing, observed a mortality benefit with prone compared with supine positioning with a modest empirical PEEP-FIO_2 strategy.[37] Concomitant proning with higher PEEP has not been tested in a large-scale trial, but effects may be synergistic.[27,38] Optimal PEEP titration with prone positioning warrants future study. Sixth, biological heterogeneity of patients with ARDS may contribute to differential therapeutic response, as suggested by post hoc analyses of other recent ARDS trials.[39,40]

Conclusions

Among patients with moderate to severe ARDS, P_{ES}-guided PEEP, compared with empirical high PEEP-FIO_2, resulted in no significant difference in death and days free from mechanical ventilation. These findings do not support P_{ES}-guided PEEP titration in ARDS.

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REFERENCES

1. Slutsky AS, Ranieri VM. Ventilator-induced lung injury. *N Engl J Med*. 2013;369(22):2126-2136. doi:10.1056/NEJMra1208707

2. Amato MB, Barbas CS, Medeiros DM, et al. Effect of a protective-ventilation strategy on mortality in the acute respiratory distress syndrome. *N Engl J Med*. 1998;338(6):347-354. doi:10.1056/NEJM199802053380602

3. Brower RG, Matthay MA, Morris A, Schoenfeld D, Thompson BT, Wheeler A; Acute Respiratory Distress Syndrome Network. Ventilation with lower tidal volumes as compared with traditional tidal volumes for acute lung injury and the acute respiratory distress syndrome. *N Engl J Med*. 2000;342(18):1301-1308. doi:10.1056/NEJM200005043421801

4. Talmor D, Sarge T, O'Donnell CR, et al. Esophageal and transpulmonary pressures in acute respiratory failure. *Crit Care Med*. 2006;34(5):1389-1394. doi:10.1097/01.CCM.0000215515.49001.A2

5. Loring SH, O'Donnell CR, Behazin N, et al. Esophageal pressures in acute lung injury: do they represent artifact or useful information about transpulmonary pressure, chest wall mechanics, and lung stress? *J Appl Physiol (1985)*. 2010;108(3):515-522. doi:10.1152/japplphysiol.00835.2009

6. Beitler JR, Majumdar R, Hubmayr RD, et al. Volume delivered during recruitment maneuver predicts lung stress in acute respiratory distress syndrome. *Crit Care Med*. 2016;44(1):91-99. doi:10.1097/CCM.0000000000001355

7. Talmor D, Sarge T, Malhotra A, et al. Mechanical ventilation guided by esophageal pressure in acute lung injury. *N Engl J Med*. 2008;359(20):2095-2104. doi:10.1056/NEJMoa0708638

8. Ferguson ND, Cook DJ, Guyatt GH, et al; OSCILLATE Trial Investigators; Canadian Critical Care Trials Group. High-frequency oscillation in early acute respiratory distress syndrome. *N Engl J Med*. 2013;368(9):795-805. doi:10.1056/NEJMoa1215554

9. Mercat A, Richard J-CM, Vielle B, et al; Expiratory Pressure (Express) Study Group. Positive end-expiratory pressure setting in adults with acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA*. 2008;299(6):646-655. doi:10.1001/jama.299.6.646

10. Meade MO, Cook DJ, Guyatt GH, et al; Lung Open Ventilation Study Investigators. Ventilation strategy using low tidal volumes, recruitment maneuvers, and high positive end-expiratory pressure for acute lung injury and acute respiratory distress syndrome: a randomized controlled trial. *JAMA*. 2008;299(6):637-645. doi:10.1001/jama.299.6.637

11. Brower RG, Lanken PN, MacIntyre N, et al; National Heart, Lung, and Blood Institute ARDS Clinical Trials Network. Higher versus lower positive end-expiratory pressures in patients with the acute respiratory distress syndrome. *N Engl J Med*. 2004;351(4):327-336. doi:10.1056/NEJMoa032193

12. Huang DT, Angus DC, Moss M, et al; Reevaluation of Systemic Early Neuromuscular Blockade Protocol Committee and the National Institutes of Health National Heart, Lung, and Blood Institute Prevention and Early Treatment of Acute Lung Injury Network Investigators. Design and rationale of the reevaluation of systemic early neuromuscular blockade trial for acute respiratory distress syndrome. *Ann Am Thorac Soc*. 2017;14(1):124-133. doi:10.1513/AnnalsATS.201608-629OT

13. Briel M, Meade M, Mercat A, et al. Higher vs lower positive end-expiratory pressure in patients with acute lung injury and acute respiratory distress syndrome: systematic review and meta-analysis. *JAMA*. 2010;303(9):865-873. doi:10.1001/jama.2010.218

14. Fish E, Novack V, Banner-Goodspeed VM, Sarge T, Loring S, Talmor D. The Esophageal Pressure-Guided Ventilation 2 (EPVent2) trial protocol: a multicentre, randomised clinical trial of mechanical ventilation guided by transpulmonary pressure. *BMJ Open*. 2014;4(9):e006356. doi:10.1136/bmjopen-2014-006356

15. Ranieri VM, Rubenfeld GD, Thompson BT, et al; ARDS Definition Task Force. Acute respiratory distress syndrome: the Berlin Definition. *JAMA*. 2012;307(23):2526-2533. doi:10.1001/jama.2012.5669

16. Finkelstein DM, Schoenfeld DA. Combining mortality and longitudinal measures in clinical trials. *Stat Med*. 1999;18(11):1341-1354. doi:10.1002/(SICI)1097-0258(19990615)18:11<1341::AID-SIM129>3.0.CO;2-7

17. Acion L, Peterson JJ, Temple S, Arndt S. Probabilistic index: an intuitive non-parametric approach to measuring the size of treatment

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effects. *Stat Med*. 2006;25(4):591-602. doi:10.1002/sim.2256

18. Kraemer HC. Correlation coefficients in medical research: from product moment correlation to the odds ratio. *Stat Methods Med Res*. 2006;15(6):525-545. doi:10.1177/0962280206070650

19. Schoenfeld DA, Bernard GR; ARDS Network. Statistical evaluation of ventilator-free days as an efficacy measure in clinical trials of treatments for acute respiratory distress syndrome. *Crit Care Med*. 2002;30(8):1772-1777. doi:10.1097/00003246-200208000-00016

20. Newcombe RG. Confidence intervals for an effect size measure based on the Mann-Whitney statistic, part 2: asymptotic methods and evaluation. *Stat Med*. 2006;25(4):559-573. doi:10.1002/sim.2324

21. Lin DY, Wei LJ, Ying Z. Checking the Cox model with cumulative sums of martingale-based residuals. *Biometrika*. 1993;80:557-572. doi:10.1093/biomet/80.3.557

22. Hodges JL, Lehmann EL. Estimates of location based on rank tests. *Ann Math Stat*. 1963;34(2):598-611. doi:10.1214/aoms/1177704172

23. Oeckler RA, Lee W-Y, Park M-G, et al. Determinants of plasma membrane wounding by deforming stress. *Am J Physiol Lung Cell Mol Physiol*. 2010;299(6):L826-L833. doi:10.1152/ajplung.00217.2010

24. Bilek AM, Dee KC, Gaver DP III. Mechanisms of surface-tension-induced epithelial cell damage in a model of pulmonary airway reopening. *J Appl Physiol (1985)*. 2003;94(2):770-783. doi:10.1152/japplphysiol.00764.2002

25. Gattinoni L, Caironi P, Cressoni M, et al. Lung recruitment in patients with the acute respiratory distress syndrome. *N Engl J Med*. 2006;354(17):1775-1786. doi:10.1056/NEJMoa052052

26. Caironi P, Cressoni M, Chiumello D, et al. Lung opening and closing during ventilation of acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2010;181(6):578-586. doi:10.1164/rccm.200905-0787OC

27. Cornejo RA, Díaz JC, Tobar EA, et al. Effects of prone positioning on lung protection in patients with acute respiratory distress syndrome. *Am J Respir Crit Care Med*. 2013;188(4):440-448. doi:10.1164/rccm.201207-1279OC

28. Cavalcanti AB, Suzumura ÉA, Laranjeira LN, et al; Writing Group for the Alveolar Recruitment for Acute Respiratory Distress Syndrome Trial (ART) Investigators. Effect of lung recruitment and titrated positive end-expiratory pressure (PEEP) vs low PEEP on mortality in patients with acute respiratory distress syndrome: a randomized clinical trial. *JAMA*. 2017;318(14):1335-1345. doi:10.1001/jama.2017.14171

29. Borges JB, Costa ELV, Suárez-Sipmann F, et al. Early inflammation mainly affects normally and poorly aerated lung in experimental ventilator-induced lung injury. *Crit Care Med*. 2014;42(4):e279-e287. doi:10.1097/CCM.0000000000000161

30. Steinberg JM, Schiller HJ, Halter JM, et al. Alveolar instability causes early ventilator-induced lung injury independent of neutrophils. *Am J Respir Crit Care Med*. 2004;169(1):57-63. doi:10.1164/rccm.200304-544OC

31. Baedorf Kassis E, Loring SH, Talmor D. Mortality and pulmonary mechanics in relation to respiratory system and transpulmonary driving pressures in ARDS. *Intensive Care Med*. 2016;42(8):1206-1213. doi:10.1007/s00134-016-4403-7

32. Gattinoni L, Pelosi P, Suter PM, Pedoto A, Vercesi P, Lissoni A. Acute respiratory distress syndrome caused by pulmonary and extrapulmonary disease: different syndromes? *Am J Respir Crit Care Med*. 1998;158(1):3-11. doi:10.1164/ajrccm.158.1.9708031

33. Terragni P, Mascia L, Fanelli V, Biondi-Zoccai G, Ranieri VM. Accuracy of esophageal pressure to assess transpulmonary pressure during mechanical ventilation. *Intensive Care Med*. 2017;43(1):142-143. doi:10.1007/s00134-016-4589-8

34. Washko GR, O'Donnell CR, Loring SH. Volume-related and volume-independent effects of posture on esophageal and transpulmonary pressures in healthy subjects. *J Appl Physiol (1985)*. 2006;100(3):753-758. doi:10.1152/japplphysiol.00697.2005

35. Pelosi P, Goldner M, McKibben A, et al. Recruitment and derecruitment during acute respiratory failure: an experimental study. *Am J Respir Crit Care Med*. 2001;164(1):122-130. doi:10.1164/ajrccm.164.1.2007010

36. Lai-Fook SJ, Rodarte JR. Pleural pressure distribution and its relationship to lung volume and interstitial pressure. *J Appl Physiol (1985)*. 1991;70(3):967-978. doi:10.1152/jappl.1991.70.3.967

37. Guérin C, Reignier J, Richard J-C, et al; PROSEVA Study Group. Prone positioning in severe acute respiratory distress syndrome. *N Engl J Med*. 2013;368(23):2159-2168. doi:10.1056/NEJMoa1214103

38. Beitler JR, Guérin C, Ayzac L, et al. PEEP titration during prone positioning for acute respiratory distress syndrome. *Crit Care*. 2015;19:436. doi:10.1186/s13054-015-1153-9

39. Calfee CS, Delucchi K, Parsons PE, Thompson BT, Ware LB, Matthay MA; NHLBI ARDS Network. Subphenotypes in acute respiratory distress syndrome: latent class analysis of data from two randomised controlled trials. *Lancet Respir Med*. 2014;2(8):611-620. doi:10.1016/S2213-2600(14)70097-9

40. Famous KR, Delucchi K, Ware LB, et al; ARDS Network. Acute respiratory distress syndrome subphenotypes respond differently to randomized fluid management strategy. *Am J Respir Crit Care Med*. 2017;195(3):331-338. doi:10.1164/rccm.201603-0645OC

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